

COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

SET C

Time: 3 Hours

Total Marks: 100

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form “prove”, “show”, or “derive” require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. For finite sets A, B (e.g. the sets of product categories two customers have purchased from), define the symmetric-difference cardinality $d(A, B) = |A \triangle B| = |A \setminus B| + |B \setminus A|$.

(a) Prove that d is a genuine metric on the collection of finite subsets of a universal set U : verify non-negativity, identity of indiscernibles, symmetry, and the triangle inequality $d(A, C) \leq d(A, B) + d(B, C)$. (*Hint: first prove the set-level inclusion $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$, then take cardinalities.*)

(b) The Jaccard *distance* used for market-basket comparison is instead $d_J(A, B) = \frac{|A \triangle B|}{|A \cup B|}$. Compute $d_J(A, B)$ for $A = \{\text{Electronics, Books, Sports}\}$ and $B = \{\text{Electronics, Sports, Ho}$ and compare it with the corresponding value of $|A \triangle B|$ from part (a). Explain in one or two sentences (no proof required) why normalizing by $|A \cup B|$ is essential for comparing basket similarity across customers with very different total basket sizes, even though it complicates the metric proof.

[8 marks]

Q2. An online store records order values (in Rs.) for $n = 9$ customers in one promotional cohort:

420, 450, 460, 480, 490, 500, 520, 550, 3800.

Compute Q_1 , the median, Q_3 , and IQR via the median-of-halves convention, and hence the Tukey fences. State which observation(s) are flagged. [4 marks]

Q3. As before, let a fraction $\varepsilon \in (0, 1)$ of n recorded order amounts be identical, coordinated glitch values equal to M (e.g. a payment-gateway bug that logs many failed retries at the same placeholder amount), with the remaining amounts confined to a bounded range negligible next to M .

(a) Derive $\mu(M)$, $\sigma(M)$, and show $z(M) \rightarrow \sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \rightarrow \infty$.

(b) Solve for the exact contamination fraction ε^* at which this limiting z -value equals 3, and state, for an e-commerce fraud-detection pipeline, the operational implication of your answer: what fraction of a batch being corrupted identically is enough to make the classical 3σ rule structurally blind to it, forever, regardless of how large the glitch value M is?

- (c) Prove that the MAD-based modified z -score instead diverges to $+\infty$ for any $\varepsilon < 0.5$, and explain why an analyst who claims “the mean/std method breaks with even a single outlier, so it is strictly worse than MAD in every respect” is conflating two different notions of breakdown; state both notions precisely and show they give different answers for this dataset.

[9 marks]

Q4. Two products are described by (Weight in grams, Price in Rs. paise): $P = (250, 45000)$, $Q = (800, 62000)$.

- (a) Compute the raw Euclidean distance and identify the dominating coordinate.
 (b) Given catalogue statistics $\mu_{\text{wt}} = 400, \sigma_{\text{wt}} = 150$ and $\mu_{\text{price}} = 50000, \sigma_{\text{price}} = 9000$, standardize both coordinates and recompute the Euclidean distance.
 (c) A “customers who bought similar products” recommender uses raw k -NN before scaling. Explain, quoting your two numerical distances, why price (recorded in paise rather than rupees) would silently dominate every recommendation regardless of physical similarity.

[6 marks]

Q5. Two products’ feature tags over {Waterproof, Wireless, Rechargeable, Foldable, Eco-friendly} are $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

- (a) Compute the Jaccard similarity and cosine similarity of x and y .
 (b) A third product listing $z = (0, 0, 0, 0, 0)$ has no tags filled in (an incomplete seller listing, not a genuine claim that the product has none of these properties). State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why silently treating z as a genuine all-absent tag vector would corrupt a tag-similarity-based product de-duplication system.

[4 marks]

Q6. Let R be the missingness indicator for a column $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

- (a) State Rubin’s formal definitions of MCAR, MAR, and MNAR.
 (b) Prove the ignorability result under MAR (with distinct parameters), showing the observed-data likelihood factors cleanly.
 (c) An online marketplace observes that sellers with the *lowest* product ratings are least likely to fill in the optional “defect rate” self-disclosure field, i.e. $P(R = 0 \mid \text{DefectRate})$ increases as the true (unrecorded) DefectRate increases. Classify this mechanism formally, and use your proof in (b)–(c) to explain concretely why no imputation using only observed listing fields can, even in principle, remove the resulting bias in average disclosed defect rate.
 (d) Suppose the marketplace additionally has an independent, always-observed automated return-rate signal correlated with true defect rate. Explain, in general terms, how conditioning on this variable could move the analysis from MNAR towards a recoverable MAR-type situation relative to the enlarged observed set, and why this shows the classification is relative to the observed variable set, not intrinsic to DefectRate itself.

[8 marks]

- Q7.** An analyst applies iterative predictive imputation to five product records over columns Weight (g), Rating (out of 5), and Price (Rs.):

Row	Weight	Rating	Price
1	300	4.2	1200
2	450	4.5	?
3	220	3.8	900
4	?	4.6	1600
5	600	4.0	1750

Missing cells are initialized with the column mean over observed values. The cyclic order is **Price** → **Weight**, using the fitted relations

$$\widehat{\text{Price}} = 2 \cdot \text{Weight} + 100 \cdot \text{Rating} - 50, \quad \widehat{\text{Weight}} = 0.25 \cdot \text{Price} - 50 \cdot \text{Rating} + 300.$$

- State the two initial fill values.
- Perform one full iteration (Row 2's Price, then Row 4's Weight), showing all substituted numbers.
- Explain, referencing the specific predictor columns involved, why a second iteration leaves both values unchanged here, and state a general condition on predictor overlap under which it would not.

[6 marks]

- Q8.** A marketplace seller-deduplication system computes pairwise similarities for three seller accounts A, B, C : $s(A, B) = 0.72$, $s(B, C) = 0.70$, $s(A, C) = 0.12$, with lower threshold 0.4 and upper threshold 0.65.

- Classify each pair.
- State the connected-components clustering of $\{A, B, C\}$ under “match” edges, and explain the transitivity problem.
- With $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}} (s(i, j) - \theta)$, $\theta = 0.5$, compute Obj for $\{A, B, C\}$ versus $\{A, B\}$, $\{C\}$, and conclude which is objectively preferable.
- For a chain of n seller accounts with adjacent similarity s_0 (just above upper threshold) and non-adjacent similarity floor f (below lower threshold), derive Obj exactly for (i) the single connected-component clustering and (ii) the clustering retaining only adjacent pairs, and prove the gap is $\Theta(n^2)$. State the operational lesson for a marketplace running seller-identity resolution across a long chain of near-duplicate accounts.

[10 marks]

- Q9.** Order values (in Rs.) for ten transactions show a strongly right-skewed pattern: 15, 20, 28, 40, 60, 110, 200, 300, 400, 500. Compute $\log_{10}$ of each to two decimal places, and state the raw ratio max/min against the transformed range, quantifying the compression achieved.

[4 marks]

- Q10.** An e-commerce team fits a Yeo–Johnson power transformer for *Order Value* and a one-hot encoder for *Payment Method* on the full dataset (train and test combined) before splitting.

- Let $n_{\text{tr}}, n_{\text{te}}$ be train/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population means of a numerical feature on each subpopulation, which may differ (e.g. test data drawn from a later, higher-inflation period). Derive the exact bias of the leaked full-data statistic relative to the train-only statistic, and state the condition under which it vanishes.

- (b) Explain formally why this non-zero bias implies any test-set metric computed downstream is optimistically biased.
- (c) A new Payment Method (e.g. a newly launched wallet) appears only at test time. State precisely how a leak-free encoder fitted on training categories only must represent this, and explain why fitting on the union of all categories seen across train and test — even though no numeric statistic is touched — is still leakage.

[8 marks]

Q11. Two candidate duplicate seller records are compared on four fields: business-name similarity = 0.9 (Jaro–Winkler), tax-ID agreement = 1, phone agreement = NaN (missing on one record), address similarity = 0.6. Weights: name 0.35, tax-ID 0.30, phone 0.20, address 0.15.

- (a) Compute the weighted-mean score using only available fields, weights renormalized.
- (b) Classify against thresholds lower = 0.5, upper = 0.8.
- (c) Explain why substituting 0 for the missing phone field instead of excluding and renormalizing would bias the decision, stating the direction of the bias.

[6 marks]

Q12. A page-load-time metric (seconds) recorded for $n = 7$ product pages: 1.2, 1.3, 1.1, 1.4, 1.2, 1.3, 9.0.

- (a) Compute the mean, standard deviation, and ordinary z -score of the value 9.0.
- (b) Compute the median, MAD, and modified z -score of 9.0.
- (c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 9.0, and connect the disagreement to your general result in Q3.

[4 marks]

Q13. For $x, y \in \mathbb{R}^d \setminus \{0\}$, recall $\cos(x, y) = \frac{x^\top y}{\|x\|_2 \|y\|_2}$.

- (a) State the Cauchy–Schwarz inequality and use it to prove rigorously that $\cos(x, y) \in [-1, 1]$ for all non-zero x, y .
- (b) Prove that $\cos(x, y) = 1$ if and only if $y = \lambda x$ for some $\lambda > 0$, and that $\cos(x, y) = -1$ if and only if $y = \lambda x$ for some $\lambda < 0$. (*Hint:* examine the equality condition of Cauchy–Schwarz, which holds iff x, y are linearly dependent.)
- (c) A product-recommendation engine treats $\cos(x, y) > 0.99$ as “effectively identical” product vectors. Using part (b), explain precisely what geometric relationship between x and y this threshold is approximating, and why cosine similarity alone cannot distinguish a product from another copy of itself sold at a different scale of feature normalization (e.g. raw counts versus per-unit-weight counts).

[8 marks]

Q14. Two blocking passes generate candidate duplicate seller pairs among $n = 7000$ standardized records. Pass 1 (same registered city and registration year) yields blocks of sizes 70, 55, 40. Pass 2 (same last-four registered-phone digits) yields blocks of sizes 9, 13, 6, 18, 11. The two candidate sets overlap in 60 pairs.

- (a) Compute the candidate pairs from each pass and their union’s size.

- (b) Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.
- (c) Ground truth contains 450 true duplicate pairs; Pass 1 alone captures 390, Pass 2 alone captures 250, both jointly capture 210. Verify via inclusion–exclusion the pair completeness of the union and state its value.
- (d) Prove $1 - (1 - p_1)(1 - p_2) \geq \max(p_1, p_2)$ for independent capture probabilities p_1, p_2 , and relate this to your answer in (c).

[6 marks]

Q15. A marketplace data-quality officer gives you eight seller-product records:

ID	Listing Age (mo.)	Package Vol. (L)	Package Wt. (kg)	Price (Rs.)	Seller Name (std.)
1	34	17.0	6.8	620	apex traders
2	29	16.5	6.0	540	apex traders
3	45	16.8	6.5	?	nova retail
4	51	17.2	7.0	1450	vk enterprises
5	38	18.0	4.2	610	gt supplies
6	60	16.6	6.3	?	d. mart
7	41	17.1	6.6	590	nova retail
8	33	16.9	6.4	605	priya goods

Among the six rows with observed Price, mean Listing Age is 37.67; among the two with missing Price, mean Listing Age is 52.5. From the other seven rows (excluding Row 5), $\mu = (\mu_V, \mu_W) = (16.94, 6.51)$ and $\Sigma = \begin{bmatrix} 0.168 & 0.076 \\ 0.076 & 0.072 \end{bmatrix}$ for (Volume, Weight). Rows 3 and 7 are suspected duplicate listings, with comparison vector: name similarity 0.95, phone NaN, city agreement 1, registration-date agreement 0, using Q11's weights/thresholds.

- (a) State, with the standard caveat about MNAR being unprovable from observed data alone, what the age-conditioned evidence does and does not establish about the missingness mechanism of Price.
- (b) Compute, by hand, the Mahalanobis distance of Row 5's $(V, W) = (18.0, 4.2)$ from μ , computing Σ^{-1} explicitly, and comment on whether Row 5 is a multivariate outlier.
- (c) Decide, via the Q11 machinery, whether Rows 3 and 7 should be merged, and construct the merged record's Price field using the course's numeric golden-record rule.
- (d) Argue formally whether deduplication should precede or follow imputation here, prove your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]